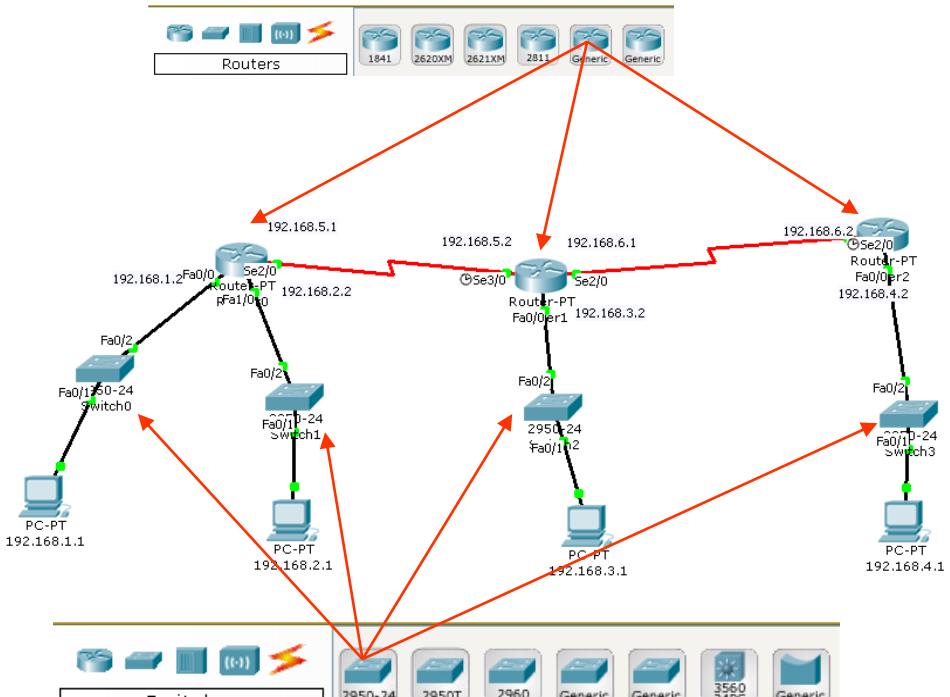
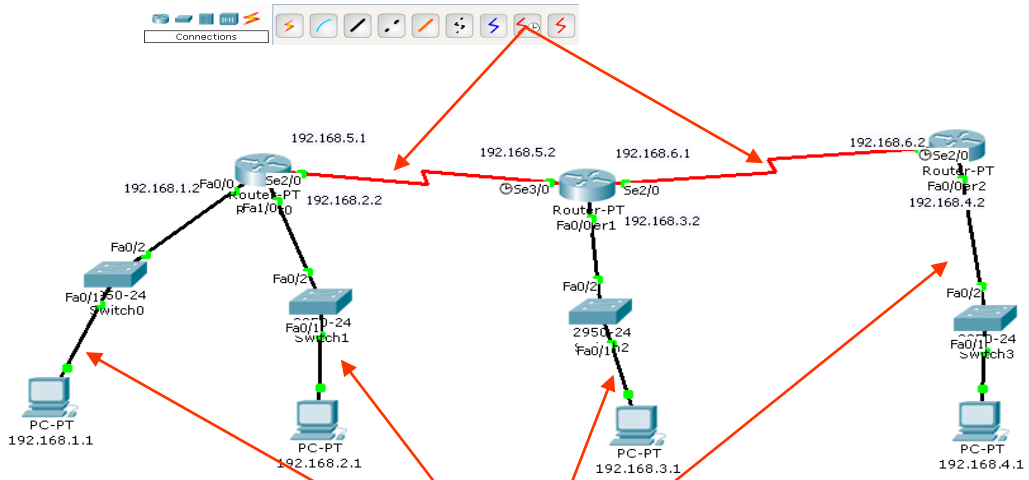
Expt-9
Routing of Packet through Complicated Network

This experiment can be considered as an extension of experiment 8. Under this experiment student will get idea about, how a router take the decision to deliver the packet to appropriate interface.

Step-1

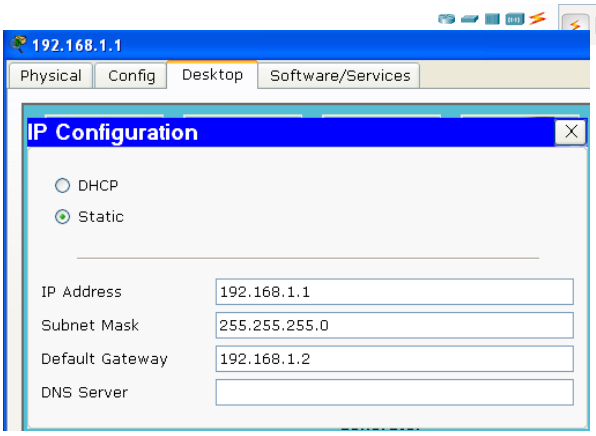
Let us implement the circuit below where the steps the depicted in the figures.

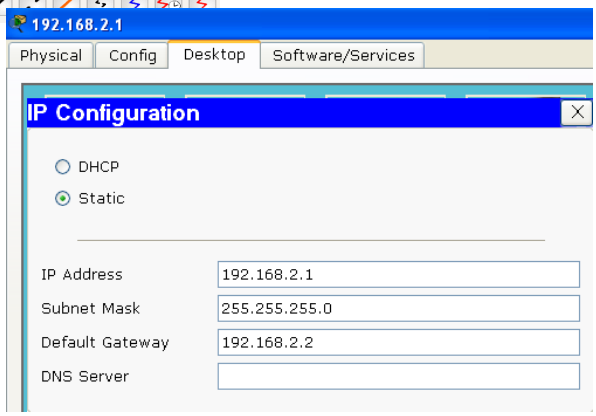


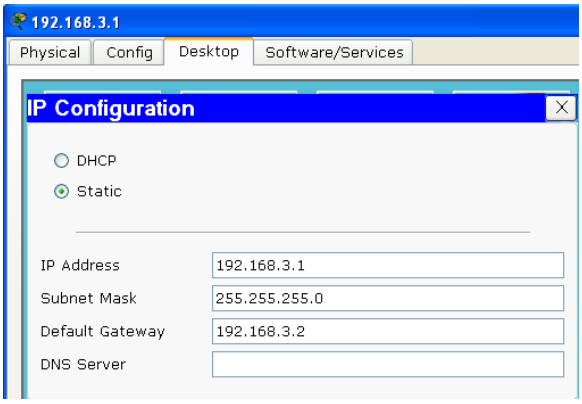


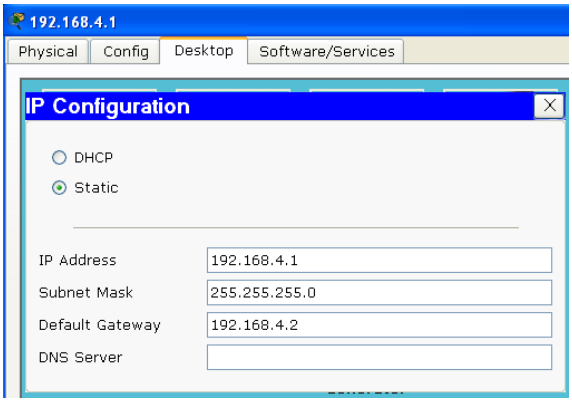
Step-2

Configure the PCs like below.









Step-3
Configure the Router-0 from CLI like:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip add 192.168.1.2 255.255.255.0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#int fa1/0
Router(config-if)#ip add 192.168.2.2 255.255.255.0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
Router(config-if)#exit
Router(config)#int s2/0
Router(config-if)#ip add 192.168.5.1 255.255.255.0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#exit
Router(config)#do wr
Building configuration…
[OK]
Router(config)#
```

Step-4
Configure the Router-1 from CLI like:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip add 192.168.3.2 255.255.255.0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#int s3/0
Router(config-if)#ip add 192.168.5.2 255.255.255.0
```

```
Router(config-if)#no shut
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
Router(config-if)#exit
Router(config)#int s2/0
Router(config-if)#ip add 192.168.6.1 255.255.255.0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#exit
Router(config)#do wr
Building configuration…
[OK]
Router(config)#
```

Step-5

Configure the Router-2 from CLI like:

```
Router-2
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int s2/0
Router(config-if)#ip add 192.168.6.2 255.255.255.0
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#int fa 0/0
Router(config-if)#ip add 192.168.4.2 255.255.255.0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#do wr
Building configuration…
[OK]
```

Step-6

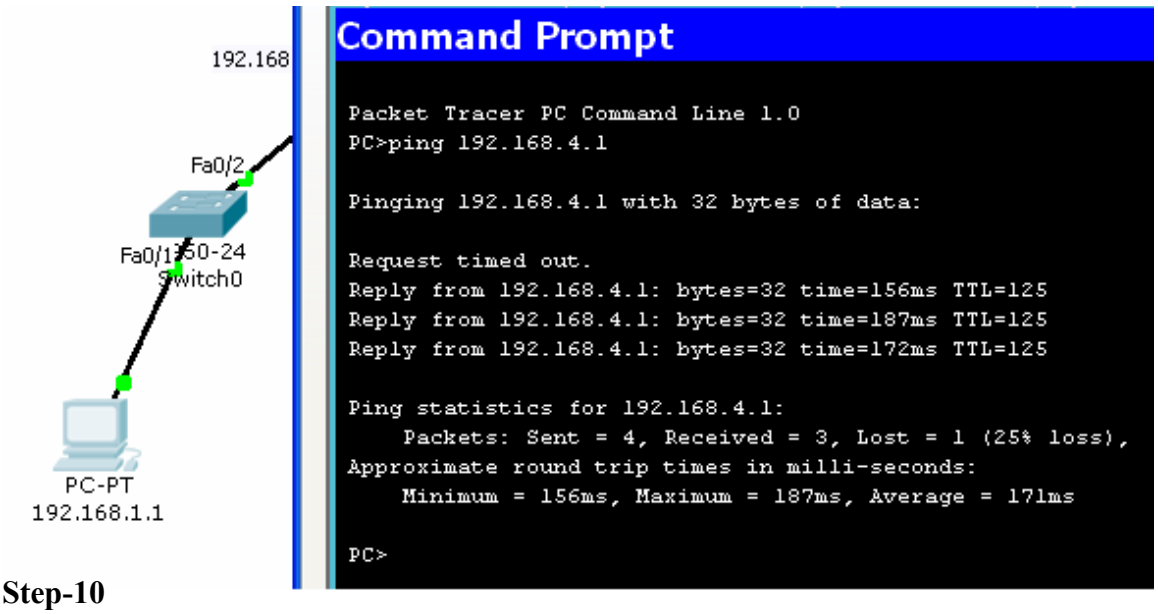
Configure the Router-0 again like:

```
Router(config)#ip route 192.168.4.0 255.255.255.0 192.168.5.2
Router(config)#ip route 192.168.3.0 255.255.255.0 192.168.5.2
Router(config)#ip route 192.168.6.0 255.255.255.0 192.168.5.2
Router(config)#do wr
Building configuration…
[OK]
Router(config)#
```

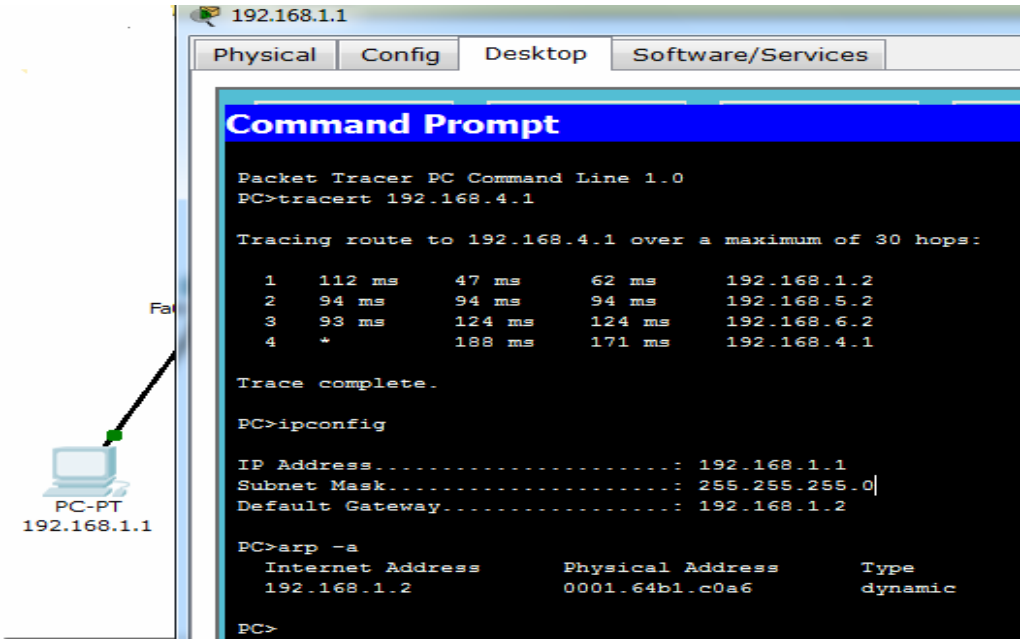
Step-7
Configure the Router-0 again like:
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.5.1
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.5.1
Router(config)#ip route 192.168.4.0 255.255.255.0 192.168.6.2
Router(config)#do wr
Building configuration…
[OK]
Router(config)#

Step-8
Configure the Router-2 again like:
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.6.1
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.6.1
Router(config)#ip route 192.168.5.0 255.255.255.0 192.168.6.1
Router(config)#ip route 192.168.3.0 255.255.255.0 192.168.6.1
Router(config)#do wr
Building configuration…
[OK]
Router(config)#

Step-9
Verify the ping command on PC of 192.168.1.1 like:

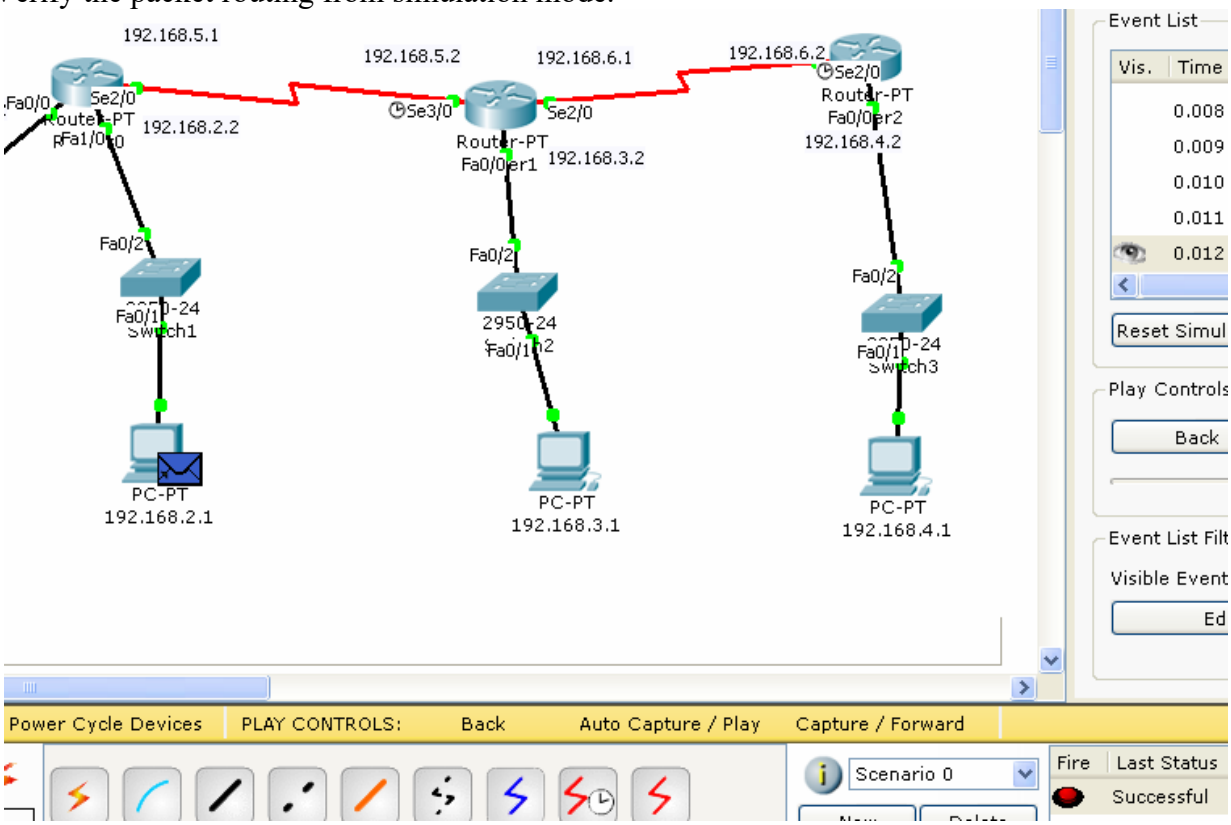


Step-10
Verify the *tracert*, *ipconfg* and *arp –a* commands like:



Step-11

Verify the packet routing from simulation mode.



Exercise

The performance of the network is 75% i.e. 25% packets are lost. How will you improve the network for 100% successful reception?